

MICHAEL LEE

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EDUCATION

Colorado School of Mines | Golden, CO Aug 2027 — Dec 2028 (expected)

M.S. in Mechanical Engineering — Focus: Thermal, Fluid & Energy Systems

Northwestern University | Evanston, IL Jun 2025

B.S. in Mechanical Engineering — Focus: Aerospace

Courses: Propulsion, Aerodynamics, Compressible Flow, Heat Transfer, CFD, Thermal Systems Design, Manufacturing Processes

PROFESSIONAL EXPERIENCE

Entegris | Colorado Springs, CO Jul 2026 – Present

Metrology Engineering Co-op

- Design and fabricate test fixtures using SolidWorks CAD and 3D printing to improve measurement repeatability; interpret CAD drawings with GD&T to develop and qualify CMM programs for tight-tolerance parts used in semiconductor packaging
- Automate measurement data analysis using Python (pandas) to parse and format spreadsheet data for Minitab statistical analysis, enabling quicker assessment of CMM program and process stability
- Train new metrology technicians on ZONE3 CMM software, including part fixturing, probe/optical measurement setup, and inspection best practices
- Work alongside manufacturing engineers and technicians to revise CNC machining SOPs for precision aluminum plates used in semiconductor EUV lithography equipment
- Improve inspection workflows by incorporating feedback from quality inspectors and metrology technicians closest to the end product; perform FMEA on proposed process changes and document in Management of Change (MOC) system

PROJECTS

Gear Reducer Transmission System Mar 2025 – Jun 2025

- Designed and integrated shaft #1 in a 3-shaft, 56 kW gearbox, reducing input speed from 2400 rpm to 337 rpm by calculating gear forces and combined stresses to guide material selection and bearing sizing
- Sized a 35 mm diameter AISI 1050 steel shaft by performing fatigue and static yield failure analyses, achieving >10 million cycles with safety factors > 1.27; selected bearings and verified L10 bearing life at operating speeds
- Built complete Siemens NX CAD shaft assembly model and detailed part drawings with standard fits and tolerances, incorporating keyways and auxiliary components (end caps, gaskets, spacers) to ensure proper manufacturing and assembly

Aerospace Testbed Design and Integration Jan 2025 – Jun 2025

- Designed a modular wing testbed for a high school summer aerospace course, defining requirements for four subsystems — avionics, wing manufacturing, wing-fuselage attachment, and launcher — enabling 30+ students to build and test custom wings
- Developed avionics subsystem using Siemens NX CAD and circuit diagrams; processed flight sensor data in Python, achieving lift and drag coefficients within 10% of predictions
- Managed 5-month project timeline using Gantt and RAM charts to assign and track responsibilities across 7-person team, resulting in on-time delivery of deliverables, including final report, bill of materials, fabrication guide, and presentation

ANSYS Fluent CFD Analysis of Fluid Mixing Feb 2025 – Mar 2025

- Simulated Rayleigh-Taylor instabilities in ANSYS Fluent using the Volume of Fluid multiphase model to study how density differences between two fluids affect the mixing interface
- Applied CFD fundamentals — meshing, boundary conditions, and verification & validation methods to ensure accuracy
- Wrote Python scripts using FFT-based analysis and sine-curve fitting to extract growth rates from simulation data
- Compared simulated mixing behavior against linear stability theory, confirming that greater density differences increase mixing growth rate and reduced error by 5.7% through solver optimization

High Power Rocket Apr 2023

- Assembled a 49.5" high-power rocket powered by an H-class motor with university's rocketry club, gaining hands-on experience with epoxy resins and fin fillets
- Executed launch and recovery in compliance with NAR and FAA regulations, earning Level 1 certification

SKILLS

CAD & Engineering: SolidWorks CAD, Siemens NX, GD&T, ANSYS (Fluent CFD, Mechanical FEA), NX FEA, Python, MATLAB

Prototyping: 3D Printing, Manual Milling, Hand & Power Tools, Electronics Soldering & Wiring